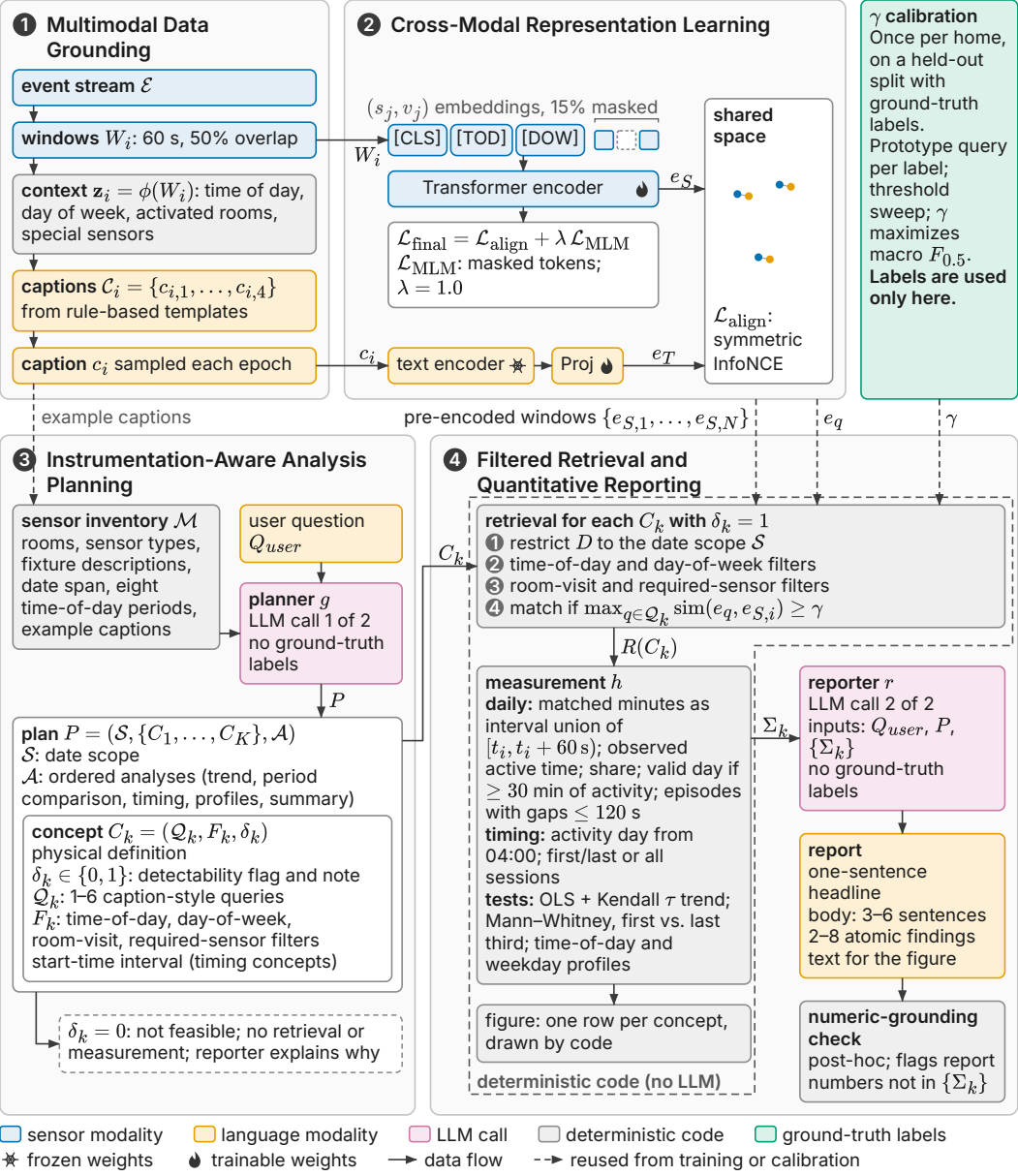

1 Multimodal Data Grounding
event stream $\mathcal{E}$
windows W_i: 60 s, 50% overlap
context $\mathbf{z}_i = \phi(W_i)$: time of day, day of week, activated rooms, special sensors
captions $\mathcal{C}_i = \{c_{i,1}, \ldots, c_{i,4}\}$ from rule-based templates
caption c_i sampled each epoch
2 Cross-Modal Representation Learning
(s_j, v_j) embeddings, 15% masked
[CLS] [TOD] [DOW]
W_i
Transformer encoder
e_S
$\mathcal{L}_{\text{final}} = \mathcal{L}_{\text{align}} + \lambda \mathcal{L}_{\text{MLM}}$
$\mathcal{L}_{\text{MLM}}$: masked tokens; $\lambda = 1.0$
shared space
$\mathcal{L}_{\text{align}}$: symmetric InfoNCE
c_i
text encoder
Proj
e_T
γ calibration
Once per home, on a held-out split with ground-truth labels. Prototype query per label; threshold sweep; γ maximizes macro $F_{0.5}$. Labels are used only here.
example captions
pre-encoded windows $\{e_{S,1}, \ldots, e_{S,N}\}$
e_q
γ
3 Instrumentation-Aware Analysis Planning
sensor inventory $\mathcal{M}$ rooms, sensor types, fixture descriptions, date span, eight time-of-day periods, example captions
user question Q_{user}
planner g LLM call 1 of 2 no ground-truth labels
P
plan $P = (\mathcal{S}, \{C_1, \ldots, C_K\}, \mathcal{A})$
$\mathcal{S}$: date scope
$\mathcal{A}$: ordered analyses (trend, period comparison, timing, profiles, summary)
concept $C_k = (\mathcal{Q}_k, F_k, \delta_k)$
physical definition
$\delta_k \in \{0, 1\}$: detectability flag and note
$\mathcal{Q}_k$: 1–6 caption-style queries
F_k: time-of-day, day-of-week, room-visit, required-sensor filters
start-time interval (timing concepts)
$\delta_k = 0$: not feasible; no retrieval or measurement; reporter explains why
4 Filtered Retrieval and Quantitative Reporting
C_k
retrieval for each C_k with $\delta_k = 1$
1 restrict D to the date scope $\mathcal{S}$
2 time-of-day and day-of-week filters
3 room-visit and required-sensor filters
4 match if $\max_{q \in \mathcal{Q}_k} \text{sim}(e_q, e_{S,i}) \geq \gamma$
$R(C_k)$
measurement h
daily: matched minutes as interval union of $[t_i, t_i + 60\,\text{s})$; observed active time; share; valid day if ≥ 30 min of activity; episodes with gaps ≤ 120 s
timing: activity day from 04:00; first/last or all sessions
tests: OLS + Kendall τ trend; Mann–Whitney, first vs. last third; time-of-day and weekday profiles
figure: one row per concept, drawn by code
deterministic code (no LLM)
Σ_k
reporter r
LLM call 2 of 2
inputs: Q_{user}, P, $\{\Sigma_k\}$
no ground-truth labels
report
one-sentence headline
body: 3–6 sentences
2–8 atomic findings
text for the figure
numeric-grounding check
post-hoc; flags report numbers not in $\{\Sigma_k\}$
sensor modality
language modality
LLM call
deterministic code
ground-truth labels
frozen weights
trainable weights
data flow
reused from training or calibration